

Notes: Kisgen & Strahan (RFS, 2010)

Do Regulations Based on Credit Ratings Affect a Firm's Cost of Capital?

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Bond, rating, and yield data	4
3.2	Rating scale and relative DBRS rating	4
3.3	Yield change outcome	4
3.4	Investment-grade boundary	5
3.5	Mechanism variables	5
4	Models and empirical specifications	5
4.1	Baseline event-study regression	5
4.2	Better-versus-worse asymmetric specification	5
4.3	Investment-grade interaction	6
4.4	Falsification designs	6
4.5	DBRS rating-behavior regressions	6
5	Identification and empirical design	6
5.1	Core identification logic	6
5.2	Why the event is useful	7
5.3	Why comparing DBRS to Moody's and S&P matters	7
5.4	Assumptions	7
5.5	Threats and how the paper addresses them	7
6	Tables and figures	8
6.1	Table 1: Sample Description	8
6.2	Table 2: Summary Statistics; Table 3: DBRS-rated Sample vs. Other S&P-rated Compustat Firms	9
6.3	Table 4: Cost of Capital Changes When DBRS Becomes an NRSRO	9
6.4	Table 5: Cost of Capital Falls for Firms Rated Better by DBRS; Table 6: Cost of Capital Falls Most for Firms Near the Investment-Grade/Speculative-Grade Cutoff	10
6.5	Table 7: Changes in Holdings and Liquidity for DBRS-rated Bonds; Table 8: Robustness Tests	11
6.6	Table 9: DBRS Ratings Converge to Other NRSROs	12
7	Short summary	13
8	Conclusion	13

8.1	Core conclusion	13
8.2	Economic intuition	13
8.3	Interpretation of the magnitude	13
8.4	Policy implications	13
8.5	Limitations and caveats	14
8.6	Takeaway	14

1 Big picture

- **Question.** Do credit ratings affect bond yields only because they contain information about default risk, or do ratings also affect yields because regulations use ratings to restrict or penalize investors' bond holdings?
- **Natural experiment.** In February 2003, the SEC designated Dominion Bond Rating Service (DBRS) as a Nationally Recognized Statistical Rating Organization (NRSRO). Before that date, DBRS ratings were visible to the market but did not have the same formal regulatory status as ratings from S&P, Moody's, and Fitch.
- **Main empirical strategy.** The paper holds DBRS ratings fixed as of January 2003 and asks whether, after DBRS becomes an NRSRO, bond yields move in the direction implied by the DBRS rating relative to Moody's and S&P.
- **Main result.** If DBRS rates a bond one notch better than Moody's and S&P, the bond's yield falls by about 39 basis points by December 2003. Around the investment-grade cutoff, the effect is about 54–55 basis points.
- **Why this is regulatory.** The effect is asymmetric: yields fall when DBRS is more favorable, but do not rise meaningfully when DBRS is less favorable. This fits regulations that rely on the best or second-best NRSRO rating. It is less consistent with a pure certification or information story.
- **Falsification.** The same design does not produce similar effects when the event is falsely moved to 2004 or 2005.
- **Mechanism evidence.** Insurance company holdings and liquidity patterns move in ways consistent with regulated investors becoming more willing or able to hold bonds that DBRS rates favorably.
- **Rating-agency behavior.** After NRSRO status, DBRS expands into more U.S. issuers and lower-grade bonds, and its ratings become more closely aligned with the incumbent NRSROs.
- **Economic takeaway.** Ratings-based regulation can lower a firm's debt cost of capital independently of default-risk information. Credit ratings are not just signals; they are also regulatory keys that affect investor demand and market liquidity.

2 Incremental contribution of the paper

- **Separates information effects from regulatory effects.** Prior work showed that ratings matter, but it was hard to know whether they matter because they reveal information or because regulation makes them binding. This paper uses DBRS's regulatory-status change to isolate the regulatory channel while holding the pre-existing ratings fixed.
- **Reverse natural experiment relative to Kliger–Sarig.** Kliger and Sarig study a change in the information content of Moody's ratings with regulatory status fixed. Kisgen and Strahan study a change in regulatory status with information content fixed.
- **Direct cost-of-capital evidence.** Instead of showing that firms with ratings have more debt or different leverage, the paper measures changes in bond yields, directly linking regulation to the cost of debt capital.
- **Asymmetry test distinguishes regulation from certification.** If SEC certification made DBRS more credible, both favorable and unfavorable DBRS ratings should matter. The paper finds mostly one-sided effects, matching regulations that reward favorable ratings more than they penalize unfavorable ones.
- **Investment-grade boundary evidence.** The paper shows that the effect is concentrated near the boundary where regulations are most binding. This provides a sharper institutional interpretation than a generic ratings-yield relation.

- **Shows rating agencies change behavior after regulatory recognition.** DBRS becomes broader and more similar to incumbents after gaining NRSRO status, suggesting that the regulatory designation can affect rating-agency strategy as well as issuer yields.

3 Data and measurement

3.1 Bond, rating, and yield data

The paper uses Bloomberg monthly data from 2001 to 2005. The key objects are bond-level yields and issuer/bond ratings from DBRS, S&P, Moody's, and Fitch.

- The sample includes nonfinancial corporate bonds with a DBRS rating and an observable yield.
- For the main event-study tests, the sample is restricted to bonds already rated by DBRS as of January 2003, one month before DBRS receives NRSRO status.
- This main sample contains 42 firms and 90 bonds. If a firm has multiple bonds, the paper includes all bonds and clusters standard errors by firm.
- Firm characteristics such as asset size and leverage are merged from Compustat.

Interpretation: The main sample restriction is central. By keeping only bonds that DBRS rated before NRSRO status, the study avoids selection from new firms seeking a DBRS rating after the SEC announcement.

3.2 Rating scale and relative DBRS rating

The paper converts letter ratings into numeric ratings:

$$AAA = 1, \quad AA+ = 2, \quad \dots$$

so that a larger number means a worse rating.

The key rating variable is:

$$DBRSMean_i = DBRS_{i,Jan2003} - \frac{Moody's_{i,Jan2003} + S\&P_{i,Jan2003}}{2}.$$

- If $DBRSMean_i < 0$, DBRS gives the bond a better rating than Moody's and S&P.
- If $DBRSMean_i = 0$, DBRS agrees with the incumbent NRSROs.
- If $DBRSMean_i > 0$, DBRS gives the bond a worse rating.

Interpretation: Because the ratings are fixed as of January 2003, variation in $DBRSMean_i$ captures the change in regulatory use of an already-known rating, not new information released by DBRS.

3.3 Yield change outcome

For each month T after the event, the dependent variable is:

$$\Delta Yield_{i,T} = Yield_{i,T} - Yield_{i,Jan2003}.$$

- The event month is February 2003, when DBRS becomes an NRSRO.
- The post-event months run from March to December 2003.
- The January 2003 yield controls for all information already impounded in bond prices before regulatory status changes.

Interpretation: The paper is asking whether the same DBRS rating becomes more value-relevant after it becomes usable in regulation.

3.4 Investment-grade boundary

The paper defines a near-investment-grade indicator:

$$NearIG_i = 1$$

if the average Moody's/S&P rating lies in the BBB–BB range.

Interpretation: This variable matters because many rating-based rules bind most sharply at the investment-grade/speculative-grade boundary. If the result is regulatory, the effect should be larger near this cutoff.

3.5 Mechanism variables

The paper also examines changes in holdings and liquidity:

- insurance company holdings from NAIC regulatory filings;
- purchases by new investors;
- TRACE turnover for a small subset of bonds;
- bond illiquidity measures based on price changes around trades.

Interpretation: These variables are not the main identification source, but they help explain why yields fall: more regulated investors can buy or hold favorably rated bonds, and liquidity may improve.

4 Models and empirical specifications

4.1 Baseline event-study regression

For each post-event month T , the paper estimates:

$$(Yield_T - Yield_{Jan2003})_i = \alpha + \beta_1(DBRSMeanRating)_{Jan2003,i} + \varepsilon_i.$$

- $DBRSMeanRating$ equals the DBRS rating minus the average Moody's/S&P rating.
- A positive β_1 means that worse DBRS ratings lead to higher yield changes, or equivalently that better DBRS ratings lead to lower yields.
- Each row of Table 4 is a separate cross-sectional regression for a different post-event month.

Interpretation: The regression uses only cross-sectional differences in how DBRS rated bonds relative to incumbent NRSROs before the event. The time-series change is the regulatory recognition of DBRS.

4.2 Better-versus-worse asymmetric specification

To distinguish regulation from certification, the paper estimates:

$$(Yield_T - Yield_{Jan})_i = \alpha + \beta_2 Better_i \times DBRSMean_i + \beta_3 Worse_i \times DBRSMean_i + \varepsilon_i.$$

- $Better_i = 1$ if DBRS rates the bond better than Moody's/S&P.
- $Worse_i = 1$ if DBRS rates the bond worse than Moody's/S&P.

Interpretation: A certification story predicts that both favorable and unfavorable DBRS ratings should matter. A regulatory story predicts a stronger effect when DBRS is favorable, because many rules use the highest or second-highest NRSRO rating.

4.3 Investment-grade interaction

The investment-grade specification adds an interaction:

$$\Delta Yield_{i,T} = \alpha + \beta_1 NearIG_i + \beta_2 Better_i \times DBRS Mean_i + \beta_3 Worse_i \times DBRS Mean_i + \beta_4 NearIG_i \times Better_i \times DBRS Mean_i + \varepsilon_i$$

Interpretation: The coefficient on the interaction tests whether favorable DBRS ratings are especially valuable for bonds close to the investment-grade cutoff.

4.4 Falsification designs

The paper reruns the baseline cross-sectional design with false event dates in 2004 and 2005:

$$(Yield_T - Yield_{JanY})_i = \alpha + \beta_1 (DBRS Mean Rating)_{JanY,i} + \varepsilon_i, \quad Y \in \{2004, 2005\}.$$

Interpretation: If the 2003 result merely reflects ordinary dynamics in split ratings or yields, similar patterns should appear around the false event dates. They do not.

4.5 DBRS rating-behavior regressions

To study whether DBRS changes behavior after NRSRO status, the paper estimates rating regressions of the form:

$$DBRS Rating_{it} = \alpha_i + \lambda_t + \rho Mean Rating_{it}^{S\&P, Moody's} + \delta Post_t \times Mean Rating_{it}^{S\&P, Moody's} + X'_{it} \gamma + u_{it}.$$

The paper also uses ordered logit specifications with issue size, Canadian issuer indicators, near-investment-grade indicators, and bond yields.

Interpretation: These tests are descriptive rather than fully causal. They show that DBRS ratings become more correlated with incumbent ratings after the agency obtains regulatory status.

5 Identification and empirical design

5.1 Core identification logic

The identification problem is that ratings normally combine information and regulation. A bond rated highly may have a lower yield because it is safer, because the rating agency has revealed information, or because regulations make the bond easier to hold.

The paper solves this problem by exploiting a regulatory-status shock:

$$DBRS \text{ rating known before February 2003} + NRSRO \text{ status granted in February 2003.}$$

The rating is held fixed, while its regulatory relevance changes.

Interpretation: This is the central advantage of the design. Any information in the January 2003 DBRS rating should already be reflected in January 2003 bond yields. The post-event yield change isolates the new regulatory usefulness of that rating.

5.2 Why the event is useful

DBRS became the fourth NRSRO in February 2003. Before this event, DBRS ratings did not count in the same way for many regulatory rules. After the event, DBRS ratings could be used by banks, insurance companies, broker-dealers, money market funds, and private contracts that reference NRSRO ratings.

- The event changes the regulatory validity of an existing rating agency.
- The main sample uses ratings fixed before the event.
- The outcome is a change in yields after the event.

Interpretation: The paper is not using a rating change. It is using a change in the legal treatment of the rating.

5.3 Why comparing DBRS to Moody's and S&P matters

If DBRS agrees with Moody's and S&P, making DBRS an NRSRO does not materially change the regulatory rating of the bond. If DBRS gives a more favorable rating, the bond may now qualify for regulations or investor mandates that depend on the best or second-best NRSRO rating.

Thus the treatment intensity is:

$$DBRS_{Jan2003} - Mean(Moody's, S\&P)_{Jan2003}.$$

Interpretation: The design is effectively a difference-in-differences in regulatory relevance: bonds with favorable DBRS ratings receive a larger regulatory improvement than bonds with DBRS ratings equal to incumbent ratings.

5.4 Assumptions

- **Predetermined ratings.** January 2003 DBRS ratings must be known and priced before the NRSRO event.
- **No other simultaneous differential shock.** Bonds with better DBRS ratings must not experience other unobserved yield shocks exactly after February 2003.
- **Regulatory relevance.** Investors and institutions must treat DBRS ratings differently after the NRSRO designation.
- **Comparability.** Bonds with different DBRS-vs-incumbent rating gaps can be compared after controlling for January yields through the yield-change outcome.

Interpretation: The paper cannot observe every investor mandate directly. It therefore triangulates the regulatory interpretation through asymmetry, investment-grade concentration, falsification tests, and holdings/liquidity evidence.

5.5 Threats and how the paper addresses them

- **Information certification.** SEC designation could make DBRS ratings more credible. The paper addresses this by showing the effect is asymmetric and concentrated where regulation is most binding.
- **Selection into DBRS coverage.** Firms might seek DBRS ratings when they expect favorable treatment. The main tests use only bonds already rated by DBRS before NRSRO status.

- **General yield trends.** The dependent variable is a yield change from January 2003, and each regression has an intercept that absorbs aggregate spread changes.
- **Spurious event timing.** The 2004 and 2005 falsification tests do not reproduce the main results.
- **Credit-quality changes.** Robustness controls include changes in other NRSRO ratings and initial rating levels.

Interpretation: The empirical design is strongest for showing that existing rating differences became priced differently after DBRS became an NRSRO. It is less able to fully measure all investor-level channels because detailed holdings are only partially observed.

6 Tables and figures

6.1 Table 1: Sample Description

Read:

- DBRS coverage grows sharply after NRSRO status: the sample rises from 46 issuers in 2002 to 165 in 2005.
- The share of Canadian issuers falls from 87% in 2001 to 51% in 2005, showing that DBRS moves beyond its Canadian base.
- DBRS expands toward lower rating categories after 2003, including more BBB and BB firms.

Detailed interpretation: Table 1 is important because it shows that NRSRO status changes DBRS’s market position. Before certification, DBRS is mainly a Canadian and high-grade agency. After certification, it becomes more like the incumbent agencies in both coverage and rating distribution.

How it fits the paper: This table supports the paper’s secondary theme: regulatory recognition does not only affect bond prices; it also reshapes the rating agency’s business model and client base.

Table 1
Sample description

Year	Number of Observations	Number of Bonds	Number of Issuers	Fraction Canadian Issuers
2001	313	57	32	87%
2002	818	113	46	78%
2003	1,534	235	73	71%
2004	3,876	539	119	57%
2005	8,223	877	165	51%
Total	14,764	1,821	435	

Before NRSRO (2001–02)		After NRSRO (2003–05)			
Distribution of DBRS-rated bonds	DBRS Ratings	S&P Ratings Distribution 2001–02	DBRS Rating		S&P Ratings Distribution 2003–05
			Existing Customers	New Customers	
AAA	5.8%	1.1%	3.4%	7.1%	0.9%
AA	7.7%	4.7%	4.0%	5.4%	3.9%
A	59.5%	20.3%	49.2%	38.3%	18.3%
BBB	26.9%	28.4%	43.4%	36.6%	30.1%
BB	0.2%	20.4%	0.0%	10.1%	22.7%
B	0.0%	18.5%	0.0%	2.4%	19.9%
CCC and worse	0.0%	6.6%	0.0%	0.2%	4.2%

This table reports the number of observations over time for which we can observe the DBRS rating, along with the distribution of the DBRS ratings and the S&P ratings for all Compustat firms across the ratings bins. The sample includes all bonds with a DBRS rating reported in Bloomberg issued by non-financial companies.

Table 1: Sample Description. The table reports DBRS coverage and rating distributions before and after DBRS becomes an NRSRO.

6.2 Table 2: Summary Statistics; Table 3: DBRS-rated Sample vs. Other S&P-rated Compustat Firms

Read:

- Table 2 shows that DBRS-rated bonds have lower average yields after 2003, while rating differences across agencies widen.
- DBRS ratings remain somewhat more favorable than the average Moody's/S&P rating.
- Table 3 shows DBRS-rated issuers are much larger than the typical S&P-rated Compustat firm and are more concentrated in regulated industries.

Detailed interpretation: These tables help describe the treated sample. The sample is not a random set of all firms; it is larger, more regulated, and connected to DBRS's historical Canadian and high-grade business. This matters because the empirical design relies on within-sample rating differences rather than comparing DBRS-rated firms with all firms.

How it fits the paper: The summary statistics justify later controls for issue size, Canadian issuers, and initial credit quality. They also highlight why the pre-event restriction is needed to avoid selection into DBRS coverage after NRSRO recognition.

Table 2
Summary statistics

	Before DBRS became an NRSRO (N = 1,225)		After DBRS became an NRSRO (N = 13,539)		P-value ¹ : (H ₀ : Pre=Post)
	Mean	Standard Deviation	Mean	Standard Deviation	
Yield to Maturity (%)	6.43	0.95	5.23	1.07	0.00
Fraction rated the same by all agencies	73.0%	–	64.0%	–	0.66
Absolute rating difference (Best – Worst)	0.30	0.53	0.52	0.83	0.77
Mean Rating (S&P & Moody's) – 1 means AAA, 2 means AA+, ...	6.30	2.30	6.96	3.15	0.01
DBRS rating – Mean Rating	–0.18	1.04	–0.28	0.86	0.06

This table reports summary statistics on yields and ratings for the sample of bonds for which we can observe the DBRS rating. The sample includes monthly data on all bonds with a DBRS rating reported in Bloomberg issued by non-financial companies, from 2001 through the end of 2005.

¹The *p*-value tests the null that the mean in the two periods is the same. The statistic is built controlling for bond-level fixed effects and clustering the data by issuer.

Table 2: Summary Statistics. Summary of yields, rating dispersion, and DBRS relative ratings before and after NRSRO status.

DBRS-rated sample vs. other S&P-rated Compustat firms

	DBRS Rated		S&P Rated	
	Mean	Standard Deviation	Mean	Standard Deviation
Leverage (total liabilities / book assets)	0.68	0.14	0.69	0.20
Asset size (MM \$)	21.5	40.5	7.3	18.9
PP&E / assets	0.07	0.06	0.06	0.06
Share mining & construction (SIC = 1)	16.7%	–	6.0%	–
Share manufacturing (SIC = 2 or 3)	29.2%	–	42.9%	–
Share regulated (SIC = 4)	46.4%	–	26.4%	–

This table reports simple summary statistics for our sample of DBRS-rated firms that are available in Compustat with all S&P-rated Compustat firms that were not rated by DBRS. Neither sample contains financial companies (SIC = 6). Data are pooled across the years 2001–2005.

Table 3: DBRS-rated Sample vs. Other S&P-rated Compustat Firms. DBRS-rated issuers are larger and more concentrated in regulated industries.

6.3 Table 4: Cost of Capital Changes When DBRS Becomes an NRSRO

Read:

- Table 4 is the baseline test. Each row is a cross-sectional regression for a post-event month.
- The 2003 coefficient rises gradually and becomes significant later in the year.
- By December 2003, a one-notch better DBRS relative rating corresponds to a 16-basis-point yield reduction.
- Falsification tests using 2004 and 2005 event dates show no similar pattern.

Detailed interpretation: The gradual timing is economically sensible. Regulated investors may not instantly rebalance portfolios, but once DBRS ratings become legally relevant, bonds with favorable DBRS ratings become more attractive. The absence of similar effects in 2004 and 2005 is crucial because it shows the result is not a generic feature of split ratings.

How it fits the paper: This is the first main evidence that regulatory recognition changes the pricing of existing credit ratings.

Table 4
Cost of capital changes when DBRS becomes an NRSRO

	Year in which DBRS becomes an NRSRO		Falsification Tests			
	2003		2004		2005	
	DBRS-Mean rating, January 2003	N (R^2)	DBRS-Mean rating, January 2004	N (R^2)	DBRS-Mean rating, January 2005	N (R^2)
One-month after	-0.04	89	-0.01	208	0.01	497
NRSRO (March)	(0.96)	0.03	(0.83)	0.01	(0.03)	0.01
Two-months after	0.02	89	-0.01	202	-0.11	494
NRSRO (April)	(1.11)	0.01	(0.11)	0.01	(1.71)	0.04
Three-months after	0.03	90	-0.01	208	-0.08	499
NRSRO (May)	(1.04)	0.01	(0.33)	0.01	(1.60)	0.02
Four-months after	0.05	90	-0.03	206	-0.03	495
NRSRO (June)	(1.21)	0.03	(0.69)	0.01	(0.79)	0.01
Five-months after	0.05	90	-0.01	204	-0.01	491
NRSRO (July)	(1.00)	0.02	(0.03)	0.01	(0.07)	0.01
Six-months after	0.07	86	-0.05	201	-0.02	495
NRSRO (August)	(1.44)	0.04	(1.41)	0.03	(0.36)	0.01
Seven-months after	0.09	86	-0.04	200	-0.01	495
NRSRO (September)	(1.44)	0.07	(0.97)	0.01	(0.12)	0.01
Eight-months after	0.11	85	-0.03	206	-0.03	499
NRSRO (October)	(1.70)	0.08	(0.85)	0.01	(0.44)	0.01
Nine-months after	0.14	85	-0.05	206	-0.03	493
NRSRO (November)	(2.34)	0.13	(1.18)	0.02	(0.42)	0.01
Ten-months after	0.16	89	-0.05	206	-0.03	499
NRSRO (December)	(2.17)	0.13	(1.00)	0.02	(0.37)	0.01

Each row in this table represents three cross-sectional regressions. We report regressions of the change in the yield to maturity on bonds after January 2003, the month before the SEC deemed DBRS an NRSRO in columns 1 & 2, controlling for the difference between the DBRS rating and the average rating (from Moody's and S&P) as of January 2003. Each regression includes an intercept that is not reported. The regressions in columns 3 & 4 and 5 & 6 report analogous cross-sectional models in 2004 and 2005, and act as falsification tests since there was no change in DBRS's regulatory status during these years. t -statistics are reported in parentheses based on residuals clustered by firm (since some firms issue more than one bond).

Table 4: Cost of Capital Changes When DBRS Becomes an NRSRO. Baseline event regressions and false-event falsification tests.

6.4 Table 5: Cost of Capital Falls for Firms Rated Better by DBRS; Table 6: Cost of Capital Falls Most for Firms Near the Investment-Grade/Speculative-Grade Cutoff

Read:

- Table 5 splits the DBRS rating gap into favorable and unfavorable DBRS ratings.
- A favorable DBRS rating is associated with a 39-basis-point yield decline by December 2003.
- An unfavorable DBRS rating does not produce a corresponding yield increase.
- Table 6 shows that the favorable-rating effect is much larger near the BBB/BB investment-grade boundary.
- The near-investment-grade favorable-rating effect is roughly 54–55 basis points.

Detailed interpretation: These tables are the paper's strongest evidence against a pure information-certification story. If SEC recognition simply made DBRS more credible, bad DBRS ratings should increase yields just as good DBRS ratings reduce yields. Instead, the effect is one-sided and concentrated exactly where investor mandates and capital rules bind most strongly.

How it fits the paper: Tables 5 and 6 convert the baseline event effect into a regulatory mechanism. Ratings matter most when they relax investor constraints.

Table 5
Cost of capital falls for firms rated better by DBRS

	DBRS-Mean rating if DBRS rates better	DBRS-Mean rating if DBRS rates worse	P-value testing that (1) = (2)	N (R ²)
	(1)	(2)		
One-month after NRSRO	-0.12 (1.21)	0.02 (0.71)	0.26	89 0.06
Two-months after NRSRO	0.01 (0.22)	0.03 (0.83)	0.86	89 0.01
Three-months after NRSRO	0.10 (1.43)	-0.03 (0.81)	0.17	90 0.04
Four-months after NRSRO	0.17 (2.07)	-0.04 (1.41)	0.04	90 0.08
Five-months after NRSRO	0.19 (2.25)	-0.06 (1.65)	0.03	90 0.08
Six-months after NRSRO	0.21 (2.80)	-0.05 (1.50)	0.01	86 0.12
Seven-months after NRSRO	0.27 (3.16)	-0.06 (2.18)	0.00	86 0.17
Eight-months after NRSRO	0.28 (2.73)	-0.03 (0.93)	0.01	85 0.15
Nine-months after NRSRO	0.30 (2.89)	0.01 (0.32)	0.02	85 0.19
Ten-months after NRSRO	0.39 (3.72)	-0.03 (0.76)	0.00	89 0.24

Each row in this table represents one cross-sectional regression. We report regressions of the change in the yield to maturity on bonds after January 2003, the month before the SEC deemed DBRS an NRSRO. We control for the difference between the DBRS rating and the average rating (from Moody's and S&P) as of January 2003. Each regression includes an intercept that is not reported. *t*-statistics are reported in parentheses based on residuals clustered by firm (since some firms issue more than one bond).

Table 5: Cost of Capital Falls for Firms Rated Better by DBRS.
The yield effect is concentrated on favorable DBRS ratings.

Table 6
Cost of capital falls most for firms near the investment-grade/speculative-grade cutoff

	Near Investment Grade	DBRS-Mean rating if DBRS rates better	DBRS-Mean rating if DBRS rates worse	Near Investment Grade * DBRS- Mean rating if DBRS rates better	N (R ²)
One-month after NRSRO	-0.35 (4.32)	0.02 (0.33)	-0.04 (1.43)	-0.30 (2.16)	89 0.21
Two-months after NRSRO	-0.35 (3.49)	0.07 (1.17)	-0.03 (1.06)	-0.17 (1.80)	89 0.22
Three-months after NRSRO	-0.27 (2.63)	0.11 (1.31)	-0.07 (2.06)	-0.06 (0.51)	90 0.15
Four-months after NRSRO	-0.20 (1.75)	0.09 (1.07)	-0.07 (2.21)	0.09 (0.69)	90 0.19
Five-months after NRSRO	0.14 (0.85)	0.08 (1.05)	-0.03 (1.00)	0.20 (1.34)	90 0.10
Six-months after NRSRO	0.14 (1.14)	0.07 (0.97)	-0.01 (0.42)	0.25 (1.85)	86 0.16
Seven-months after NRSRO	0.02 (0.14)	0.10 (1.42)	-0.06 (1.89)	0.28 (2.20)	86 0.25
Eight-months after NRSRO	-0.07 (0.41)	0.08 (1.04)	-0.03 (1.00)	0.30 (2.04)	85 0.25
Nine-months after NRSRO	-0.07 (0.49)	0.11 (1.57)	0.01 (0.49)	0.30 (2.18)	85 0.29
Ten-months after NRSRO	-0.01 (0.01)	0.16 (2.28)	-0.01 (0.31)	0.37 (2.65)	89 0.32

Each row in this table represents one cross-sectional regression. We report regressions of the change in the yield to maturity on bonds after January 2003, the month before the SEC deemed DBRS an NRSRO. We control for the difference between the DBRS rating and the average rating (from Moody's and S&P) as of January 2003. Near-investment-grade bonds are those with an average rating by Moody's & S&P in the BBB-BB range. Each regression includes an intercept that is not reported. *t*-statistics are reported in parentheses based on residuals clustered by firm (since some firms issue more than one bond).

Table 6: Cost of Capital Falls Most for Firms Near the Investment-Grade/Speculative-Grade Cutoff. The regulatory effect is largest near BBB/BB.

6.5 Table 7: Changes in Holdings and Liquidity for DBRS-rated Bonds; Table 8: Robustness Tests

Read:

- Table 7 shows larger insurance-company holdings for bonds rated better by DBRS and evidence of new investor purchases.
- TRACE-based turnover and illiquidity measures are limited, but they are directionally consistent with improved liquidity.
- Table 8 shows the main result survives including Fitch ratings, controls for initial rating, changes in other ratings, issue size, Canadian issuers, and alternative yield-change scaling.

Detailed interpretation: Table 7 is mechanism evidence, not the core causal test. It helps explain why yields fall: the pool of eligible or willing regulated investors expands. Table 8 shows that the effect is not simply an artifact of other rating changes, issue size, or Canadian issuer status.

How it fits the paper: These tables strengthen the interpretation that the yield decline reflects a demand/liquidity channel created by regulatory recognition.

Table 7: Changes in holdings and liquidity for DBRS-rated bonds

	Bonds Rated Better by DBRS	Bonds Rated the Same	Bonds Rated the Worse by DBRS	Correlation with DBRS ratings differential (P-value)
12/2002 Holdings as a fraction of issue size, median bond	(1) 4.79%	(2) 2.25%	0.00%	—
12/2003 Holdings as a fraction of issue size, median bond	5.93%	2.35%	0.00%	—
Change in holdings, median bond	0.8%	0.4%	0.00%	−0.02 (0.840)
Amount purchased by new investors, median bond	0.91%	0.58%	0.00%	−0.21 (0.047)
Change in Turnover, median bond ¹	−4.3%	−6.1%	—	−0.45 (0.047)
Change in Illiquidity, median bond	−0.27	−0.09	—	0.35 (0.12)

This table describes aggregate holdings by all U.S. Life and Property & Casualty insurance companies as of the end of 2002 and the end of 2003, from regulatory reports filed with the National Association of Insurance Commissioners (NAIC). Data on turnover and illiquidity are constructed from TRACE using trades from July–December 2002 with those made from July–December 2003. Turnover equals monthly volume divided by issue size. Bond illiquidity equals the negative of the correlation between price changes on consecutive trades, following Bao, Pan, and Wang (2008).

¹ Subsample of 10 bonds with trading volume available in TRACE.

Table 7: Changes in Holdings and Liquidity for DBRS-rated Bonds. Favorably rated bonds attract more investor demand and possibly better liquidity.

Table 8: Robustness tests

	Baseline model: Table 6, final row	Include Fitch when available	Add Control Variables	Log of Yield Change
Near investment grade indicator (BBB or BB)	−0.01 (0.01)	0.02 (0.08)	0.02 (0.09)	0.002 (0.06)
DBRS-mean rating if DBRS rates better	0.16 (2.28)	0.16 (2.32)	0.10 (0.97)	0.024 (1.97)
DBRS-mean rating if DBRS rates worse	−0.01 (0.31)	−0.01 (0.39)	0.08 (0.81)	−0.007 (1.29)
Near investment grade *	0.37 (2.65)	0.43 (2.59)	0.42 (2.52)	0.055 (2.36)
DBRS-mean rating if DBRS rates better	—	—	—	0.061 (2.27)
Mean level of rating as of January 2003	—	—	−0.04 (2.27)	— (1.69)
Change in rating, January 2003 to December 2003	—	—	−0.11 (0.74)	— (0.29)
Log of issue size	—	—	0.01 (0.37)	— (0.04)
Indicator for Canadian issuers	—	—	−0.61 (3.23)	— (3.39)
N	89	89	86	89
R ²	0.32	0.32	0.50	0.26

Each column in this table represents one cross-sectional regression, where the dependent variable measures the change in the yield from January 2003 to December 2003. The first column reproduces that last row from Table 6. We control for the difference between the DBRS rating and the average rating (from Moody's and S&P) as of January 2003. Each regression includes an intercept that is not reported. *t*-statistics are reported in parentheses based on residuals clustered by firm (since some firms issue more than one bond).

Table 8: Robustness Tests. The baseline near-investment-grade result is stable across alternative specifications.

6.6 Table 9: DBRS Ratings Converge to Other NRSROs

Read:

- Table 9 studies DBRS rating behavior before and after NRSRO status.
- DBRS ratings become more closely linked to Moody's and S&P after NRSRO recognition.
- Before NRSRO status, issue size and bond yield predict relatively more favorable DBRS ratings.
- After NRSRO status, the special relation between DBRS ratings and issue size or yields weakens.

Detailed interpretation: This table is descriptive but economically useful. It suggests that before gaining regulatory status, DBRS may have had stronger incentives to attract issuers by rating large or high-yielding bonds more favorably. Once it becomes an NRSRO, reputational and regulatory credibility become more important, and DBRS ratings converge toward the incumbents.

How it fits the paper: Table 9 closes the loop: regulatory status affects not only prices but also the rating agency's strategy and rating standards.

Table 9: DBRS ratings converge to other NRSROs

	Only Bonds issued and rated by DBRS before they become an NRSRO			All Bonds		
Mean rating (S&P & Moody's) — 1 means AAA, 2 means AA+, ...	0.52 (2.46)	2.07 (6.72)	2.20 (7.22)	0.56 (3.58)	2.31 (8.01)	2 (8)
Log of issue size	—	−0.93 (3.81)	−0.90 (4.09)	—	−0.98 (4.45)	0 (4)
Indicator for Canadian issuers	—	−1.17 (1.78)	−1.23 (1.76)	—	−1.33 (1.80)	— (1)
Indicator for bonds near investment grade	—	0.13 (0.22)	0.17 (0.27)	—	−0.67 (1.07)	−1 (4)
Bond yield	—	—	−0.66 (2.62)	—	—	−1 (4)
Indicator for DBRS post NRSRO	−0.49 (2.52)	−1.48 (1.54)	−4.60 (3.51)	−0.50 (2.50)	−4.65 (2.71)	— (4)
Mean rating * Post NRSRO	0.07 (2.82)	0.06 (0.41)	−0.03 (0.18)	0.07 (2.74)	0.01 (0.03)	— (4)
Log of issue size * Post NRSRO	—	0.20 (1.53)	0.18 (1.54)	—	0.71 (2.33)	1 (4)
Indicator for Canadian issuers * Post NRSRO	—	−0.23 (0.52)	−0.40 (0.58)	—	0.75 (0.99)	1 (4)
Indicator for bonds near investment grade * Post NRSRO	—	−0.02 (0.04)	−0.11 (0.15)	—	0.32 (0.52)	1 (4)
Bond yield * Post NRSRO	—	—	0.60 (2.46)	—	—	1 (4)
P-value for Chi ² -test that issue size coefficient = 0 after NRSRO	—	0.00	0.00	—	0.36	1 (4)
P-value for Chi ² -test that Bond Yield coefficient = 0 after NRSRO	—	—	0.73	—	—	1 (4)
Estimation Method		Fixed Effects OLS	Ordered Logit		Fixed Effects OLS	Ordered Logit
N		4,270	0.48		14,058	0.55
R ² (within for fixed effects OLS, pseudo for ordered logit)		0.38	0.48		0.41	0.55

This table reports regressions of the DBRS rating on the average rating by Moody's and S&P along with issue characteristics. Each rating is converted into a number (e.g., AAA = 1, AA+ = 2, etc.), so an increase in the dependent variable means DBRS rates the firm worse. All of the regressors are measured as of the preceding month. We report each model's fixed-effects OLS and also using an ordered logit (w/o the bond-level fixed effects). Also, we report the model for all of our data and for the subset of bonds rated by DBRS both before and after the change in their status. *t*-statistics are reported in parentheses based on residuals clustered by firm (since some firms issue more than one bond).

Table 9: DBRS Ratings Converge to Other NRSROs. Rating regressions show DBRS becomes more aligned with incumbent agencies after NRSRO status.

7 Short summary

- The paper studies whether ratings-based regulation affects corporate bond yields.
- The natural experiment is the SEC's February 2003 recognition of DBRS as an NRSRO.
- The key design holds DBRS ratings fixed as of January 2003, so post-event yield changes cannot be explained by new DBRS information.
- Bonds with favorable DBRS ratings experience lower yields after DBRS becomes an NRSRO.
- The effect reaches about 39 basis points by December 2003 for a one-notch favorable DBRS rating.
- The effect is asymmetric: favorable DBRS ratings lower yields; unfavorable DBRS ratings do not symmetrically raise yields.
- The effect is strongest near the investment-grade boundary, where rating-based rules are most binding.
- False event years in 2004 and 2005 do not reproduce the result.
- Holdings and liquidity evidence suggest that regulated investor demand is a plausible channel.
- DBRS expands coverage and becomes more similar to Moody's and S&P after NRSRO status.
- The incremental contribution is to isolate the regulatory value of ratings from their information value.

8 Conclusion

8.1 Core conclusion

Kisgen and Strahan show that regulations based on credit ratings affect firms' cost of debt capital. The DBRS natural experiment is powerful because it changes the regulatory status of ratings without changing the ratings themselves. Once DBRS becomes an NRSRO, bonds with favorable DBRS ratings become cheaper to finance, especially near the investment-grade boundary.

8.2 Economic intuition

Ratings are not merely information. They also determine which investors can hold bonds, how much capital regulated institutions must hold, and whether bonds fit within investment mandates. When a rating becomes regulatory-relevant, investor demand can shift even if default risk is unchanged. This is why a pre-existing DBRS rating can suddenly matter more after February 2003.

8.3 Interpretation of the magnitude

The 39-basis-point main effect and roughly 55-basis-point investment-grade effect are economically large. Around the investment-grade cutoff, these magnitudes represent a meaningful share of the spread between investment-grade and speculative-grade bonds. The implication is that losing or gaining regulatory eligibility can change financing costs by much more than default-risk information alone would suggest.

8.4 Policy implications

The paper provides evidence that outsourcing regulatory rules to rating agencies affects real capital costs. This matters for debates over NRSRO designation, regulatory reliance on ratings, and competition among rating agencies. A regulatory designation can create value for issuers and for the rating agency, but it can also distort incentives and investor demand.

8.5 Limitations and caveats

The paper does not observe every institutional investor mandate or all bond-level trading before TRACE coverage. The rating-agency behavior results are more descriptive than causal. The sample is relatively small because the main identification requires bonds already rated by DBRS before February 2003. Nevertheless, the event design, falsification tests, asymmetry, and investment-grade concentration form a coherent regulatory interpretation.

8.6 Takeaway

The lasting lesson is that ratings are institutional objects, not just statistical forecasts of default. When law and regulation attach consequences to ratings, a rating agency's status can directly affect firms' financing costs.